Atrey Desai

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Google Scholar

Education

University of Maryland, College Park — *Exp. Graduation: May 2027*
B.S. in Computer Science (Honors) & B.A. in Linguistics (Minor in Korean Studies) — * – in progress
Advisors: Prof. Rachel Rudinger & Prof. Jordan Boyd-Graber
Coursework: Natural Language Processing (Grad), Commonsense Reasoning (Grad), Machine Learning, Data Science, Parallel Computing*, Computer Systems, Syntax, Phonetics, Psycholinguistics, Intermediate Korean.

Experience

Learn Prompting — *May 2025 – present*
Member of Technical Staff (San Francisco, CA)
- Researching robustness of AI safety judges against CBRNE-related adversarial content.
- Running Hackaprompt, the world's largest red-teaming hackathon, overseeing user engagement and technical challenges.

University of Maryland, College Park — *May 2024 – present*
Undergrad Researcher, CLIP Lab
Advisors: Prof. Rachel Rudinger & Prof. Jordan Boyd-Graber
- Validated robustness of LLM-generated MCQs for unintended artifacts to assess if questions are solvable without full context, providing a framework that improves the reliability of synthetic data for benchmark evaluation.
- Architected an adversarial benchmark to test VLM capabilities in detecting out-of-context (OOC) video-based misinformation on social media based on multimodal clues and user interactions.
- Used causal interpretability techniques to find LMs need significantly more data than humans to learn general representations of syntactic structure, and are more sensitive to item and construction-level variation.

The University of Texas at Arlington — *Feb 2024 – present*
Visiting Researcher, ACL2 Lab & NSF
Advisor: Prof. Kenny Zhu
- Designed AniVoice-cat, a dataset of 26,000+ annotated cat vocalizations from 250+ hours of video and identified 57 unique cat phones, establishing a foundational resource for computational research in non-human communication.
- Improved transcription pipeline using PANNs and HuBERT models to 96% accuracy with 93% top-5 accuracy in action recognition, setting a new state-of-the-art for automated animal vocalization analysis.

University of Maryland, College Park — *Dec 2023 – Aug 2024*
Researcher, FIRE Sustainability Analytics Lab
Advisor: Prof. Thanicha Ruangmas
- Developed Python pipeline for environmental impact assessments of U.S. emissions, enabling more efficient policy.
- Drafted a framework to guide evidence-based policymaking on climate restoration strategies.

Brown University — *Dec 2020 – June 2023*
Researcher, Reinforcement Learning at Brown Group
Advisor: Prof. Michael Littman
- Developed a custom RL environment that empowered non-technical users to programmatically solve complex tasks by defining reward functions and specifying agent behavior, reducing task setup time.
- Published and presented research at two top-tier workshops (AAAI, RLDM), demonstrating how human-readable interfaces enable fine-grained control during inference and improve AI-human interaction in robotics.

Publications

Preprints

* – equal contribution

4. **BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks**.
   Nishant Balepur, Bhavya Rajasekaran, Jane Oh, Michael Xie, **Atrey Desai**, . . . , Jordan Boyd-Graber.
   (Under Review at ACL Rolling Review).

3. **Filling in the Mechanisms: How do LMs Learn Filler-Gap Dependencies under Developmental Constraints?**.
   Atrey Desai and Sathvik Nair.
   (Under Review at ACL Rolling Review), *Oral at Texas Linguistics Society (**TLS**), 2026*.

2. **Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers**.
   Nishant Balepur, **Atrey Desai** and Rachel Rudinger.
   arXiv:2510.07761 (Under Review at ACL Rolling Review).

1. **A Preview of Computational Animal Linguistics**.
   Atrey Desai, Tirza Panunto, Lindsay Pike, Theron S. Wang, Tuan M. Dang, Hridayesh Lekhak and Kenny Q. Zhu.
   (Under Review at Computational Linguistics).

Conference Publications

2. **Language Models Generate Multiple-Choice Questions with Artifacts**.
   Atrey Desai, Nishant Balepur and Rachel Rudinger.
   *Mid-Atlantic Student Colloquium on Speech, Language and Learning (**MASC-SLL**), 2025.*

1. **Reinforcement Learning As End-User Trigger-Action Programming**.
   Chace Hayhurst, Hyojae Park, **Atrey Desai**, Suheidy De Los Santos and Michael Littman.
   *AAAI Conference on Artificial Intelligence (**AAAI**) IML Workshop, 2022.*
   *Reinforcement Learning and Decision Making (**RLDM**), 2022.*

Selected Awards and Honors

- **SPIRE Research Grant** ($3,000) *2025*
- **Omicron Delta Kappa Top 10 Freshman** *2024*
- **CMSC & ARHU Dean's List** *2023 – 2025*
- **UMD President's Scholarship** ($50,000) *2023*
- **NMSC National Merit Scholarship** ($4,000) *2023*
- **Catherine Yang Scholarship** ($1,000) *2023*

Talks

- **Filler-Gap Dependencies under Developmental Constraints in LMs**
  - The University of Texas at Austin *Feb 2026 (Upcoming)*
- **Adaptor Grammars and Neural Networks for Feline Lexical Discovery**
  - The University of Texas at Arlington *July 2024*
  - University of Maryland, College Park *(adapted)* *Nov 2024*

Professional Responsibilities

- **Subreviewer**, ACL Rolling Review (ARR) *2025*
- **Member**, University Research Advisory Board *2025 – present*
- **Vice Chair**, Computer Science Department Council *2025 – present*
  Appointed by faculty and student body to serve as vice chair; represent 4,200+ CS undergraduates.
- **Senior Member**, FIRE Student Leadership Council *2024 – present*
  Represented 1000+ peers, ran events & workshops, and reformed class curricula.
- **University Ambassador** *2024 – present*
  Represented the CS Dept. & CMNS College at admissions events and to official guests.
- **Mentorship**
  - **Office of Undergraduate Research:** Juan Cortés, Kemisola Benson, Vivian Akpala *2025*
  - **Technica Mentoring:** Savya Miriyala, Tanya Grover, Jessica Ononye, Nakshatra Hiray *2024*
  - **Technica Hackathon:** Volunteer and mentor *Oct 2024*
  - **MSET Robotics Workshops:** Organizer and curriculum designer *2020 – 2022*

Skills

- **Languages:** Python, Java, JavaScript/HTML/CSS, R, MATLAB.
- **Libraries/Frameworks:** Huggingface (Datasets, Transformers), NLTK, PyTorch, Selenium, BeautifulSoup.
- **Tools:** Git, Docker, GCP, Google Vertex AI, VS Code.
- **Natural Languages:** English (Native), Gujarati (Native), Spanish (Intermediate), Korean (Beginner).